Step 1: Install Libraries

```
!pip install nltk spacy
!python -m spacy download en_core_web_sm
```

```
Requirement already satisfied: nltk in /usr/local/lib/python3.12/dist-packages (3.9.1)
Requirement already satisfied: spacy in /usr/local/lib/python3.12/dist-packages (3.8.11)
Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages (from nltk) (8.3.1)
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Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from nltk) (4.67.1)
Requirement already satisfied: spacy-legacy<3.1.0,>=3.0.11 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.0.12)
Requirement already satisfied: spacy-loggers<2.0.0,>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.0.5)
Requirement already satisfied: murmurhash<1.1.0,>=0.28.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.0.15)
Requirement already satisfied: cymem<2.1.0,>=2.0.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.13)
Requirement already satisfied: preshed<3.1.0,>=3.0.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.0.12)
Requirement already satisfied: thinc<8.4.0,>=8.3.4 in /usr/local/lib/python3.12/dist-packages (from spacy) (8.3.10)
Requirement already satisfied: wasabi<1.2.0,>=0.9.1 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.1.3)
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Requirement already satisfied: catalogue<2.1.0,>=2.0.6 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.10)
Requirement already satisfied: weasel<0.5.0,>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.4.3)
Requirement already satisfied: typer-slim<1.0.0,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.21.1)
Requirement already satisfied: numpy>=1.19.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.2)
Requirement already satisfied: requests<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.32.4)
Requirement already satisfied: pydantic!=1.8,!=1.8.1,<3.0.0,>=1.7.4 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.1
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.1.6)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from spacy) (75.2.0)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (25.0)
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!=1.8.1,<3.
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!=1.8.1,<3.0
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Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0->spacy) (3.
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0->spac
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0->spac
Requirement already satisfied: blis<1.4.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.4->spacy) (
Requirement already satisfied: confection<1.0.0,>=0.0.1 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.4->sp
Requirement already satisfied: cloudpathlib<1.0.0,>=0.7.0 in /usr/local/lib/python3.12/dist-packages (from weasel<0.5.0,>=0.4.2-
Requirement already satisfied: smart-open<8.0.0,>=5.2.1 in /usr/local/lib/python3.12/dist-packages (from weasel<0.5.0,>=0.4.2->s
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->spacy) (3.0.3)
Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packages (from smart-open<8.0.0,>=5.2.1->weasel<0.5.0,>=0
Collecting en-core-web-sm==3.8.0
  Downloading https://github.com/explosion/spacy-models/releases/download/en_core_web_sm-3.8.0/en_core_web_sm-3.8.0-py3-none-any
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 12.8/12.8 MB 68.8 MB/s eta 0:00:00
✓ Download and installation successful
You can now load the package via spacy.load('en_core_web_sm')
⚠ Restart to reload dependencies
If you are in a Jupyter or Colab notebook, you may need to restart Python in
order to load all the package's dependencies. You can do this by selecting the
'Restart kernel' or 'Restart runtime' option.
```

Step 2: Import Libraries

```
import nltk
import spacy
from nltk.tokenize import sent_tokenize, word_tokenize
from nltk.stem import PorterStemmer
```

```
nltk.download('punkt')
nltk.download('punkt_tab')
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt_tab.zip.
True
```

Step 3: Load a Simplified Medical Text Corpus

Double-click (or enter) to edit

```
medical_text = """
Diabetes mellitus is a chronic metabolic disorder characterized by high blood sugar levels.
```

```
Patients with diabetes often require insulin therapy to manage glucose levels.
Early diagnosis and proper treatment can prevent severe complications.
"""


print(medical_text)
```

```
Diabetes mellitus is a chronic metabolic disorder characterized by high blood sugar levels.
Patients with diabetes often require insulin therapy to manage glucose levels.
Early diagnosis and proper treatment can prevent severe complications.
```

Step 4: Sentence Tokenization (NLTK)

```
sentences = sent_tokenize(medical_text)

print("Sentences:")
for s in sentences:
    print(s)
```

```
Sentences:

Diabetes mellitus is a chronic metabolic disorder characterized by high blood sugar levels.
Patients with diabetes often require insulin therapy to manage glucose levels.
Early diagnosis and proper treatment can prevent severe complications.
```

Double-click (or enter) to edit

Step 5: Word Tokenization (NLTK)

```
words = word_tokenize(medical_text)
print("Words:")
print(words)
```

```
Words:
['Diabetes', 'mellitus', 'is', 'a', 'chronic', 'metabolic', 'disorder', 'characterized', 'by', 'high', 'blood', 'sugar', 'levels
```

Step 6: Stemming Medical Terms (NLTK)

```
stemmer = PorterStemmer()
stemmed_words = [stemmer.stem(word) for word in words if word.isalpha()]


print("Stemmed Words:")
print(stemmed_words)
```

```
Stemmed Words:
['diabet', 'mellitu', 'is', 'a', 'chronic', 'metabol', 'disord', 'character', 'by', 'high', 'blood', 'sugar', 'level', 'patient'
```

step 7 : Load spaCy Model

```
nlp = spacy.load("en_core_web_sm")
doc = nlp(medical_text)
```

Step 8: Lemmatization Using spaCy

```
lemmatized_words = [token.lemma_ for token in doc if token.is_alpha]


print("Lemmatized Words:")
print(lemmatized_words)
```

```
Lemmatized Words:
['diabetes', 'mellitus', 'be', 'a', 'chronic', 'metabolic', 'disorder', 'characterize', 'by', 'high', 'blood', 'sugar', 'level',
```

Step 9: Compare Stemming vs Lemmatization

```
comparison = list(zip(stemmed_words[:10], lemmatized_words[:10]))
comparison
```

```
[('diabet', 'diabetes'),
 ('mellitu', 'mellitus'),
 ('is', 'be'),
 ('a', 'a'),
 ('chronic', 'chronic'),
 ('metabol', 'metabolic'),
 ('disord', 'disorder'),
 ('character', 'characterize'),
 ('by', 'by'),
 ('high', 'high')]
```